Signature:

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

End-of-Module Exam: 2026 Spring Semester

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Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX08-WIN.XX / Data Science Introduction
Exam date / time:	26.08.2026 / 15:30 – 16:30
Duration:	60 Minutes
Total number of pages:	2
Permissible resources:	Closed book
Calculator:	no calculator
Dictionary:	Yes

General information:

- Use a black or blue pen (invisible ink is not allowed) to write your answers.
- All options of a question (i.e. A–D) can be ticked.
- ONLY tick ONE option per question in the table.
- Scoring: each correctly answered question = 1 point.
- Write your answers only on the answer sheet provided; rough work on question pages is not scored.
- Switch off your mobile phone and put it in your bag for the duration of the examination.
- If you want to write a draft, use the back of the question pages. Only the answer sheet counts.
- Please remain seated until all exams have been collected.

Marked by:	Total points:	Grade:

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Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1	100%		
		(Grade on cover sheet)	
Module grade (Evento)			⇐ Rounded ⇐

Marking Table

No.	Task	Points	Corrector's, initials
1	MC written Exam	30 Points	hitz
	Total	30 Points	

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Answer sheet

ONLY the answer sheet will be scored. All options of a question (i.e. A–D) can be ticked.
ONLY tick ONE option per question in this table.
Scoring: Each correctly answered question is worth **1 point**.
Example: For question 0, C should be ticked, then it would look like this.

Question	Answer	Question	Answer
1.	A B C D	16.	A B C D
2.	A B C D	17.	A B C D
3.	A B C D	18.	A B C D
4.	A B C D	19.	A B C D
5.	A B C D	20.	A B C D
6.	A B C D	21.	A B C D
7.	A B C D	22.	A B C D
8.	A B C D	23.	A B C D
9.	A B C D	24.	A B C D
10.	A B C D	25.	A B C D
11.	A B C D	26.	A B C D
12.	A B C D	27.	A B C D
13.	A B C D	28.	A B C D
14.	A B C D	29.	A B C D
15.	A B C D	30.	A B C D

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MC Data Science Introduction (30 Points)

1. A consultant wants to simulate how inventory levels and ordering delays cause oscillations over time. Which approach fits best? (1)
   A. The Viable System Model.
   B. The SMOTE algorithm.
   C. System Dynamics.
   D. A confusion matrix.

2. Which single statement about Ashby-style adaptation is correct? (1)
   A. Adaptation means permanently freezing the system's variety.
   B. A system adapts by matching the environment's variety, e.g. through amplification or attenuation.
   C. Adaptation requires ignoring the environment.
   D. If environmental variety exceeds capacity, the system automatically thrives.

3. Which statement is NOT correct regarding the Data Science distinction? (1)
   A. Analytics is one component within Data Science.
   B. Big Data and Data Science mean exactly the same thing.
   C. Big Data refers primarily to data traits and infrastructure.
   D. Data Science is a broad, transdisciplinary discipline.

4. Which statement about real-time adaptive modeling is NOT correct? (1)
   A. Concept drift is irrelevant to adaptive models.
   B. Model-based reinforcement learning builds an environment model to plan ahead.
   C. CPS event streams provide fast, continuous feedback.
   D. Stream/online learning updates the model as new data arrives.

5. A model scores 98% on the training set but only 70% on the test set. This most likely indicates... (1)
   A. underfitting.
   B. high bias.
   C. overfitting / high variance.
   D. perfect generalization.

6. In human-centric systems, the main legitimate purpose of a dashboard is to... (1)
   - A. combat data overload and make internal processes visible.
   - B. manipulate stakeholders into a decision.
   - C. replace all analysis.
   - D. hide data from users.

7. Which single statement about complexity is correct? (1)
   - A. CPS data is always IID and perfectly structured.
   - B. Complexity is merely complicatedness with many individual parts.
   - C. Complexity involves many interdependent variables whose cause and effect are non-obvious.
   - D. You can always gain full overview of and control over a complex system.

8. Which statement about class weighting is NOT correct? (1)
   - A. It helps prevent the majority class from dominating.
   - B. It is useful for imbalanced problems such as fraud detection.
   - C. It modifies the training loss.
   - D. It physically duplicates or deletes data rows.

9. Which statement about target-oriented action is NOT correct? (1)
   - A. The closed loop culminates in a concrete action (sense-think-act).
   - B. Start with a measurable goal.
   - C. Align stakeholders on the same target.
   - D. It is best to start data-first and let goals emerge later.

10. Explainable AI (XAI) primarily aims to... (1)
    - A. hide the model's internal logic.
    - B. explain why a model made a decision, supporting user trust.
    - C. maximize predictive accuracy only.
    - D. remove human oversight.

11. Which cyber-layer functions typically support CPS control? (1)
    - A. Complex Event Processing (CEP) and Statistical Process Control (SPC).
    - B. Only manual rules written by operators.
    - C. Only static spreadsheets.
    - D. None - control needs no data.

12. Which single statement about Stafford Beer's VSM is correct? (1)
    - A. It excludes feedback by design.
    - B. It is a recursive, cybernetics-based structure whose subsystems monitor and regulate one another.
    - C. It is identical to a static organisational chart.
    - D. It is a linear plan with no recursion.

13. Which statement about class imbalance is NOT correct? (1)
    - A. Resampling and class weighting can help.
    - B. SMOTE interpolates new minority samples.
    - C. High accuracy always means the minority class is well predicted.
    - D. The minority class is often the important one.

14. Which statement about uncertainty is NOT correct? (1)
    - A. Scenario thinking helps under uncertainty.
    - B. Data may exist, yet outcomes remain unpredictable.
    - C. Uncertainty differs from complexity and volatility.
    - D. Uncertainty means a total absence of data.

15. Unsupervised learning... (1)
    - A. finds hidden structure or groupings in unlabeled data.
    - B. requires labeled data.
    - C. predicts known labels for new data.
    - D. always learns from rewards and penalties.

16. 'Fatal undercomplexity' refers to... (1)
    - A. having too much controller variety.
    - B. an irrelevant theoretical concept.
    - C. control mechanisms that are insufficient for the situation's complexity.
    - D. a perfectly matched controller and system.

17. Which single statement about System Dynamics is correct? (1)
    - A. Stocks are rates of change and flows are accumulations.
    - B. Balancing loops counteract change while reinforcing loops amplify it.
    - C. Reinforcing loops always stabilise a system.
    - D. It assumes strictly linear cause and effect.

18. Which statement about the smart factory is NOT correct? (1)
    - A. It is a purely offline reporting system.
    - B. MES bridge machines and ERP systems.
    - C. It integrates physical and cyber layers.
    - D. Predictive maintenance reduces unplanned downtime.

19. Which statement about model performance evaluation is NOT correct? (1)
    - A. Accuracy can be misleading under class imbalance.
    - B. High training accuracy removes the need for evaluation.
    - C. In CRISP-DM, evaluation aligns with a predefined design and expert feedback.
    - D. Precision is correct-among-predicted-positives; recall is found-among-actual-positives.

20. Which of the following is an unsupervised clustering algorithm? (1)
    A. Linear regression
    B. A decision-tree classifier
    C. K-means
    D. Logistic regression

21. A 'Customer 360' view unifying data across systems for strategic insight is an example of... (1)
    A. overfitting.
    B. a target-/subject-centric strategy.
    C. data-first drifting.
    D. a data swamp.

22. Which single statement about clustering and network analysis is correct? (1)
    A. Clustering studies edges while network analysis forms centroids.
    B. Network analysis can only be done on labeled data.
    C. Both require fully labeled training data.
    D. Both are exploratory/unsupervised in spirit; clustering forms groups and network analysis maps connections.

23. Which statement about Big Data hype is NOT correct? (1)
    A. Much of big data is unstructured.
    B. Data lakes need metadata to remain usable.
    C. Volume alone does not guarantee value.
    D. Bigger data is always better regardless of quality.

24. Which statement about EDA is NOT correct? (1)
    A. EDA precedes and guides the modeling steps.
    B. Data profiling analyses content, structure and quality.
    C. EDA steps may safely use test-set information during training.
    D. Anomaly detection can flag unusual values early.

25. Which of the following is a common MISINTERPRETATION of Ashby's Law? (1)
    A. You must always simply add more complexity to your own system.
    B. It is a guiding principle of cybernetics.
    C. The controller must match the system's variety.
    D. Attenuating the environment's variety is a valid response.

26. In VUCA environments fixed models tend to fail, so the recommended response is... (1)
    A. ignoring the data.
    B. generic one-size-fits-all solutions.
    C. rigid long-term plans.
    D. adaptive, feedback-based decision-making (systems thinking, sense-think-act).

27. Which single statement about eager versus lazy learners is correct? (1)
    - A. Decision trees are lazy learners.
    - B. KNN is an eager learner with very fast predictions.
    - C. Eager learners store all instances and predict slowly.
    - D. Lazy learners (e.g., KNN) defer generalization to query time and use more memory.

28. Which statement about bias is NOT correct? (1)
    - A. AI automation removes all bias.
    - B. Survivorship bias ignores failures.
    - C. Confirmation bias favours data supporting prior beliefs.
    - D. Bias mitigation includes documenting assumptions.

29. Which statement about VSM early warning is NOT correct? (1)
    - A. Incomplete system knowledge can cause a control breakdown (a feedback design flaw).
    - B. Feedback becomes unnecessary once a system is deployed.
    - C. SPC and anomaly detection support early warning in data-rich CPS.
    - D. Anti-oscillatory loops help detect instability early.

30. Data Science is best described as... (1)
    - A. a transdisciplinary field (math/statistics, computer science/IT, domain knowledge, social science).
    - B. only domain knowledge.
    - C. only software engineering.
    - D. purely applied statistics.